

Mathematical Explanation of Bayesian Clustering Functions

1 Core Probability Functions

1.1 Multivariate Student- t Log Density: `log_mvt`

Computes the log probability density of a multivariate Student- t distribution:

$$\begin{aligned} \log p(\mathbf{x}|\nu, \boldsymbol{\mu}, \boldsymbol{\Sigma}) = & \log \Gamma\left(\frac{\nu + d}{2}\right) - \log \Gamma\left(\frac{\nu}{2}\right) - \frac{d}{2} \log(\nu\pi) \\ & - \frac{1}{2} \log |\boldsymbol{\Sigma}| - \frac{\nu + d}{2} \log \left(1 + \frac{(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})}{\nu}\right) \end{aligned} \quad (1)$$

where:

- $\mathbf{x} \in \mathbb{R}^d$ is the observation vector
- $\nu > 0$ is the degrees of freedom parameter
- $\boldsymbol{\mu} \in \mathbb{R}^d$ is the location parameter
- $\boldsymbol{\Sigma} \in \mathbb{R}^{d \times d}$ is the positive definite scale matrix
- $(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})$ is the Mahalanobis distance

The Mahalanobis distance measures how far $\mathbf{x}$ is from $\boldsymbol{\mu}$ in units of $\boldsymbol{\Sigma}$. This distribution has heavier tails than a Gaussian, making it robust to outliers.

2 Bayesian Clustering Algorithms

2.1 Pitman-Yor Process Mixture: `gibbs_pitmanyor_2d_collapsed`

Implements **collapsed Gibbs sampling** for a Pitman-Yor Process (PYP) mixture model with Gaussian likelihood.

2.1.1 Model Structure

The observations $\mathbf{y}_1, \dots, \mathbf{y}_n \in \mathbb{R}^d$ come from unknown clusters with Gaussian distributions. The cluster parameters (mean $\boldsymbol{\mu}_j$, covariance $\boldsymbol{\Sigma}_j$) are integrated out using Normal-Inverse-Wishart (NIW) conjugacy, yielding a predictive Student- t distribution for each cluster.

2.1.2 Pitman-Yor Prior on Partitions

The Pitman-Yor process is a generalization of the Dirichlet process that allows for more flexible cluster size distributions. The probability of assigning observation i to an existing cluster j is:

$$P(c_i = j \mid \text{rest}) \propto (n_j - d) \cdot p(\mathbf{y}_i \mid \mathbf{y}_j) \quad (2)$$

The probability of assigning observation i to a new cluster is:

$$P(c_i = \text{new} \mid \text{rest}) \propto (\alpha + d \cdot k) \cdot p(\mathbf{y}_i \mid \text{prior}) \quad (3)$$

where:

- c_i denotes the cluster assignment of observation i
- n_j is the number of observations currently in cluster j (excluding i)
- $d \in [0, 1)$ is the discount parameter, which encourages more balanced cluster sizes
- $\alpha > 0$ is the concentration parameter, controlling the tendency to create new clusters
- k is the current number of clusters
- $p(\mathbf{y}_i \mid \mathbf{y}_j)$ is the predictive likelihood of $\mathbf{y}_i$ given the data in cluster j
- $p(\mathbf{y}_i \mid \text{prior})$ is the prior predictive likelihood

2.1.3 Collapsed Gibbs Sampling

Instead of sampling cluster parameters $(\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)$ explicitly, the algorithm uses NIW posterior updates to compute predictive distributions directly. This marginalization improves mixing and efficiency.

For a cluster j with data $\mathbf{y}_j = \{\mathbf{y}_{i_1}, \dots, \mathbf{y}_{i_{n_j}}\}$, the NIW posterior hyperparameters are:

$$\kappa_n = \kappa_0 + n_j \quad (4)$$

$$\nu_n = \nu_0 + n_j \quad (5)$$

$$\mathbf{m}_n = \frac{\kappa_0 \mathbf{m}_0 + n_j \bar{\mathbf{y}}_j}{\kappa_n} \quad (6)$$

$$\boldsymbol{\Lambda}_n = \boldsymbol{\Lambda}_0 + \mathbf{S}_j + \frac{\kappa_0 n_j}{\kappa_n} (\bar{\mathbf{y}}_j - \mathbf{m}_0)(\bar{\mathbf{y}}_j - \mathbf{m}_0)^T \quad (7)$$

where:

- $\bar{\mathbf{y}}_j = \frac{1}{n_j} \sum_{i \in j} \mathbf{y}_i$ is the cluster mean
- $\mathbf{S}_j = \sum_{i \in j} (\mathbf{y}_i - \bar{\mathbf{y}}_j)(\mathbf{y}_i - \bar{\mathbf{y}}_j)^T$ is the within-cluster scatter matrix
- $\mathbf{m}_0, \kappa_0, \nu_0, \mathbf{\Lambda}_0$ are the NIW prior hyperparameters

The predictive distribution for a new observation is then Student- t with parameters:

$$\nu_{\text{pred}} = \nu_n - d + 1 \quad (8)$$

$$\boldsymbol{\mu}_{\text{pred}} = \mathbf{m}_n \quad (9)$$

$$\boldsymbol{\Sigma}_{\text{pred}} = \frac{\kappa_n + 1}{\kappa_n \nu_{\text{pred}}} \mathbf{\Lambda}_n \quad (10)$$

2.2 Product Partition Model with Covariates: gibbs_ppmx_pitmanyor_2d_collap

Extends the Pitman-Yor model by incorporating covariate information to guide clustering. This is known as a Product Partition Model with Covariates (PPMx).

2.2.1 Model Framework

The model considers two types of variables:

- Response variables $\mathbf{y}_i \in \mathbb{R}^{D_y}$ (e.g., physical coordinates like latitude-longitude)
- Covariate variables $\mathbf{x}_i \in \mathbb{R}^{D_x}$ (e.g., demographic or environmental features)

Clustering is performed on $\mathbf{y}$, but the partition prior is modulated by covariate similarity in $\mathbf{x}$ through a cohesion function $g(\mathbf{x}_{\text{cluster}})$.

2.2.2 Modified Assignment Probability

The probability of assigning observation i to an existing cluster j is:

$$P(c_i = j \mid \text{rest}) \propto (n_j - d) \cdot p(\mathbf{y}_i \mid \mathbf{y}_j) \cdot \left[\frac{g(\mathbf{x}_j \cup \{\mathbf{x}_i\})}{g(\mathbf{x}_j)} \right]^\eta \quad (11)$$

The probability of assigning observation i to a new cluster is:

$$P(c_i = \text{new} \mid \text{rest}) \propto (\alpha + d \cdot k) \cdot p(\mathbf{y}_i \mid \text{prior}) \cdot [g(\{\mathbf{x}_i\})]^\eta \quad (12)$$

where:

- $g(\mathbf{x}_{\text{cluster}})$ is the NIW marginal likelihood of covariates in the cluster, measuring how coherent the covariates are
- $\eta \geq 0$ is the strength of covariate influence ($\eta = 0$ reduces to standard Pitman-Yor)
- $\mathbf{x}_j$ denotes the set of covariates for observations in cluster j

2.2.3 Cohesion Function

The cohesion function $g(\mathbf{x}_{\text{cluster}})$ is computed as the NIW marginal likelihood:

$$g(\mathbf{x}_{\text{cluster}}) = p(\mathbf{x}_{\text{cluster}} \mid \text{NIW}) = \int p(\mathbf{x}_{\text{cluster}} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}) \cdot p(\boldsymbol{\mu}, \boldsymbol{\Sigma} \mid \text{NIW}) d\boldsymbol{\mu} d\boldsymbol{\Sigma} \quad (13)$$

This integral has a closed form (see Section 2.4).

2.2.4 Intuition

If observations have similar covariates $\mathbf{x}$, they are more likely to be clustered together even if their response values $\mathbf{y}$ are somewhat different. This allows the model to borrow strength from covariate information when the response space has weak signal.

2.3 Helper Function: Predictive Student- t Parameters

The function `pred_t_params` computes predictive Student- t parameters using NIW conjugate updates.

2.3.1 NIW Posterior Updates

Given n_j observations in a cluster with sufficient statistics $\sum \mathbf{v}_i$ and $\sum \mathbf{v}_i \mathbf{v}_i^T$, the NIW posterior hyperparameters are:

$$\bar{\mathbf{v}} = \frac{1}{n_j} \sum_{i=1}^{n_j} \mathbf{v}_i \quad (14)$$

$$\mathbf{S} = \sum_{i=1}^{n_j} (\mathbf{v}_i - \bar{\mathbf{v}})(\mathbf{v}_i - \bar{\mathbf{v}})^T = \sum_{i=1}^{n_j} \mathbf{v}_i \mathbf{v}_i^T - n_j \bar{\mathbf{v}} \bar{\mathbf{v}}^T \quad (15)$$

$$\kappa_n = \kappa_0 + n_j \quad (16)$$

$$\nu_n = \nu_0 + n_j \quad (17)$$

$$\mathbf{m}_n = \frac{\kappa_0 \mathbf{m}_0 + n_j \bar{\mathbf{v}}}{\kappa_n} \quad (18)$$

$$\boldsymbol{\Lambda}_n = \boldsymbol{\Lambda}_0 + \mathbf{S} + \frac{\kappa_0 n_j}{\kappa_n} (\bar{\mathbf{v}} - \mathbf{m}_0)(\bar{\mathbf{v}} - \mathbf{m}_0)^T \quad (19)$$

The predictive distribution for a new observation is Student- t with:

$$\nu_{\text{pred}} = \nu_n - D + 1 \quad (20)$$

$$\boldsymbol{\mu}_{\text{pred}} = \mathbf{m}_n \quad (21)$$

$$\boldsymbol{\Sigma}_{\text{pred}} = \frac{\kappa_n + 1}{\kappa_n \nu_{\text{pred}}} \boldsymbol{\Lambda}_n \quad (22)$$

where D is the dimension of the data.

2.4 Helper Function: NIW Marginal Likelihood

The function `log_niw_marginal` computes the marginal likelihood under a Normal-Inverse-Wishart model:

$$\log p(\mathbf{X}_{\text{cluster}} \mid \text{NIW}) = \log \int p(\mathbf{X}_{\text{cluster}} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}) \cdot p(\boldsymbol{\mu}, \boldsymbol{\Sigma} \mid \text{NIW}) d\boldsymbol{\mu} d\boldsymbol{\Sigma} \quad (23)$$

This has a closed form involving the multivariate gamma function:

$$\log \Gamma_p(a) = \frac{p(p-1)}{4} \log \pi + \sum_{j=1}^p \log \Gamma \left(a + \frac{1-j}{2} \right) \quad (24)$$

The full log marginal likelihood for n_j observations is:

$$\begin{aligned} \log p(\mathbf{X}_{\text{cluster}} \mid \text{NIW}) = & -\frac{n_j D}{2} \log(2\pi) + \frac{D}{2} (\log \kappa_0 - \log \kappa_n) \\ & + \log \Gamma_D \left(\frac{\nu_n}{2} \right) - \log \Gamma_D \left(\frac{\nu_0}{2} \right) \\ & + \frac{\nu_0}{2} \log |\boldsymbol{\Lambda}_0| - \frac{\nu_n}{2} \log |\boldsymbol{\Lambda}_n| \end{aligned} \quad (25)$$

where D is the dimension of $\mathbf{x}$, and $\kappa_n, \nu_n, \boldsymbol{\Lambda}_n$ are the posterior hyperparameters after observing the cluster data.

Purpose: This measures how well the covariate data $\mathbf{x}$ in a cluster fits a multivariate Gaussian model.

3 Evaluation Functions

3.1 Posterior Similarity Matrix: `posterior_similarity_matrix`

Computes the co-clustering probability matrix $\mathbf{P}$ where:

$$P_{ij} = \frac{1}{M} \sum_{m=1}^M \mathbb{1}(c_i^{(m)} = c_j^{(m)}) \quad (26)$$

This averages over M post-burn-in MCMC samples. Entry (i, j) gives the posterior probability that observations i and j belong to the same cluster.

Properties:

- $P_{ii} = 1$ for all i (diagonal elements)
- $P_{ij} = P_{ji}$ (symmetric)
- $0 \leq P_{ij} \leq 1$ (probability)

This matrix provides a complete summary of clustering uncertainty across the posterior distribution.

3.2 Binder Loss Point Estimate: `binder_ls`

Finds the partition that minimizes the Binder loss function:

$$\mathcal{L}(\mathbf{c}) = \sum_{i < j} [\mathbb{1}(c_i = c_j) - P_{ij}]^2 \quad (27)$$

where $\mathbf{c} = (c_1, \dots, c_n)$ is a partition and $\mathbf{P}$ is the posterior similarity matrix.

Procedure:

1. Compute the posterior similarity matrix $\mathbf{P}$ from the MCMC samples
2. For each sampled partition $\mathbf{c}^{(m)}$ in the post-burn-in chain:
 - (a) Construct the adjacency matrix $\mathbf{A}^{(m)}$ where $A_{ij}^{(m)} = \mathbb{1}(c_i^{(m)} = c_j^{(m)})$
 - (b) Calculate the loss: $\mathcal{L}(\mathbf{c}^{(m)}) = \sum_{i < j} (A_{ij}^{(m)} - P_{ij})^2$
3. Return the partition with minimum loss

This selects the MCMC sample that is closest to the posterior mean clustering (represented by $\mathbf{P}$), providing a single representative partition.

3.3 Partition Entropy: `calculate_entropy`

Computes the Shannon entropy of the cluster size distribution:

$$H = - \sum_{j=1}^k \frac{n_j}{n} \log_2 \left(\frac{n_j}{n} \right) \quad (28)$$

where:

- k is the number of clusters in the partition
- n_j is the size of cluster j
- $n = \sum_{j=1}^k n_j$ is the total number of observations

Interpretation:

- High entropy: many small clusters (more diverse partition)
- Low entropy: few large clusters (less diverse partition)
- Maximum entropy: $\log_2 n$ when all observations are in separate clusters
- Minimum entropy: 0 when all observations are in one cluster

This measure quantifies the uncertainty or diversity of the clustering structure.

4 Simulation Functions

The simulation functions generate synthetic datasets with known ground truth for testing the clustering algorithms under different scenarios.

4.1 `simulate_two_clusters_2d`

Generates $n_1 + n_2$ observations from two well-separated 2D Gaussian clusters:

$$\mathbf{y}_i \sim \mathcal{N}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}), \quad i = 1, \dots, n_1 \quad (29)$$

$$\mathbf{y}_i \sim \mathcal{N}(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}), \quad i = n_1 + 1, \dots, n_1 + n_2 \quad (30)$$

with $\boldsymbol{\mu}_1 = (-3, -3)^T$, $\boldsymbol{\mu}_2 = (3, 3)^T$, and $\boldsymbol{\Sigma} = \mathbf{I}_2$.

4.2 `simulate_covariates`

Generates covariates $\mathbf{x}_i \in \mathbb{R}^2$ that are informative about the true cluster structure:

$$\mathbf{x}_i \sim \mathcal{N}((-1, 0)^T, 0.36\mathbf{I}_2), \quad \text{if } c_i = 0 \quad (31)$$

$$\mathbf{x}_i \sim \mathcal{N}((1, 0)^T, 0.36\mathbf{I}_2), \quad \text{if } c_i = 1 \quad (32)$$

4.3 `simulate_overlap_physical_distinct_covariates`

Creates a challenging scenario where:

- Physical space ($\mathbf{y}$): nearly identical distributions with strong overlap

$$\mathbf{y}_i \sim \mathcal{N}((0, 0)^T, \boldsymbol{\Sigma}_{\text{phys}}), \quad i = 1, \dots, n_1 \quad (33)$$

$$\mathbf{y}_i \sim \mathcal{N}((0.3, 0.3)^T, \boldsymbol{\Sigma}_{\text{phys}}), \quad i = n_1 + 1, \dots, n_1 + n_2 \quad (34)$$

where $\boldsymbol{\Sigma}_{\text{phys}} = \begin{pmatrix} 1.0 & 0.6 \\ 0.6 & 1.0 \end{pmatrix}$

- Covariate space ($\mathbf{x}$): clearly distinct distributions

$$\mathbf{x}_i \sim \mathcal{N}((-2, 0)^T, 0.25\mathbf{I}_2), \quad \text{if } c_i = 0 \quad (35)$$

$$\mathbf{x}_i \sim \mathcal{N}((2, 0)^T, 0.25\mathbf{I}_2), \quad \text{if } c_i = 1 \quad (36)$$

This tests the PPMx model's ability to leverage covariate information when the response space provides weak signal for clustering.